UU-N-1650A
August 29, 1973
SUPERSEDING
Fed. Spec. UU-N-1650
May 10, 1972

FEDERAL SPECIFICATION

NAPKINS, TABLE, PAPER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The specification covers white, single-ply, 2-ply, and 3-ply, creped or embossed napkins.

1.2 Classification. Paper napkins in accordance with this specification shall be of one grade.

1.2.1 Types.

Type I - Eighth-folded, 3-ply napkin.
Type II - Folded, single-ply napkin.
  Class 1 - Junior dispenser napkin.
  Class 2 - Senior dispenser napkin.
  Class 3 - Quarter-folded napkin.
Type III- Eighth-folded, 2-ply napkin.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

RR-D-500 - Dispenser, Paper Napkin.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-up.
PPP-B-1163 - Box, Corrugated Fiberboard, High Compression Strength, Weather-Resistant, Wax-Resin Impregnated.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

PSC 8540

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Military Specification:

MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical Association of the Pulp and Paper Industry (TAPPI) Standards:

T 402 - Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets and Related Products.
T 403 - Bursting Strength of Paper.
T 404 - Tensile Breaking Strength of Paper and Paperboard.
T 410 - Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 432 - Water Absorption of Bibulous Paper.
T 452 - Brightness of Paper.
T 456 - Wet Tensile Breaking Strength of Paper and Paperboard.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Comparison sample. A comparison sample shall be furnished with the invitation for bids to show workmanship (see 3.8) required, and delivered napkins shall be at least equal to the comparison sample in these respects; in all other respects the specification shall apply (see 6.2).

3.2 Material. Paper napkins shall be made of cellulosic, fibrous material which may include reclaimed fibers and shall conform to the physical requirements of table I.

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TABLE I. Physical requirements 1/

Characteristic	Type I	Type II	Type III
Plies	3	1	2
Basis weight (24 x 36-500) pounds, minimum	30	12.5	20
Dry tensile strength CMD, minimum	-	0.10 kilogram (kg) per 15 millimeters (mm)	-
Both directions, average	-	0.184 kg/15 mm	-
Wet tensile strength CMD, mimimum	0.04 kg/15 mm	-	0.025 kg/15 mm
Both directions, average	0.056 kg/15 mm	-	0.042 kg/15 mm
Absorption time, 0.01 milliliter (ml) water, maximum seconds	100	100	100
Brightness, percent, minimum	82	76	76

1/ Requirements are based on napkins, as received, i.e., single, two, or three ply.

3.3 Size.

3.3.1 Type I napkins. The area of each type I single sheet of napkin shall be not less than 272 square inches, with neither dimension less than 16 inches. Napkins must be square cut, trimmed clean with even edges.

3.3.2 Type II napkins. The area of each junior dispenser napkin shall be not less than 104 square inches. The width shall be not less than 7-7/8 inches. The length shall be not less than 13 inches. The area of each senior dispenser napkin or quarter-folded napkin shall be not less than 164 square inches. Neither dimension shall be less than 12-1/2 inches. Dispenser type napkins shall be folded to comply with the following range of dimensions: width 3-3/8 to 3-5/8 inches (includes all tolerances), length 6-5/8 to 6-7/8 inches (includes all tolerances). Napkins shall be trimmed square on four sides with clean, even edges.

3.3.3 Type III napkins. The area of each type III single sheet of napkin shall be not less than 217 square inches, with neither dimension less than 14-3/4 inches. Napkins must be square cut, trimmed clean with even edges.

3.4 Finish. Types I and III napkins shall be embossed clearly with a firm ply adhesion and the design must be such as to emboss all four sides of the border. The center may be plain or either partially or entirely embossed. Type II napkins shall be creped or embossed.

3.5 Plies. Type I napkins shall be 3-ply. Type II napkins shall be single ply. Type III napkins shall be 2-ply.

3.6 Odor. The napkins shall have no disagreeable odor, either wet or dry.

3.7 Dispensing operation. Dispenser napkins (type II, class 1 and 2) shall fit and feed satisfactorily from the napkin dispenser specified in RR-D-500.

3.8 Workmanship. The napkins shall be uniformly made and edges shall be well finished. Types I and III shall be soft to the touch and lint free. All napkins shall be free from defects with respect to color, cleanliness, finish, holes, tears, breaks, wrinkles, and folding.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Sampling for inspection and test. Sampling for inspection and test shall be performed in accordance with MIL-STD-105, except where otherwise indicated. For sampling purposes, a lot shall consist of the same type and class of napkins offered for inspection at one time.

4.2 Inspection of components and materials. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, examined and tested in accordance with this specification or referenced subsidiary specifications and standards.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. Examination of the end item shall be made for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth in 4.3.1.5.

4.3.1.1 Examination of preparation for delivery. An examination shall be made to determine whether markings, materials, and workmanship comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed and closed. An equal number of shipping containers fully prepared for delivery shall be examined for closure defects. Defects shall be scored as listed below.

Examine	Defects
Marking (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence or method of application.
Material	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps; loose or inadequate sealing, strapping, or stapling.
Contents	Number of intermediate packages per shipping container is less than specified.

4.3.1.2 Examination of the end item for average count. The sample unit for this examination is the package randomly selected from each shipping container selected in accordance with 4.3.1.1. The average number of napkins per package, of all units examined, shall be not less than the quantity specified.

4.3.1.3 Examination of the end item for visual defects. The napkins shall be examined for the defects listed below. The sample unit shall be one napkin and shall be selected from the shipping containers examined under 4.3.1.1. Not more than five napkins shall be selected from any one package.

Examine	Defects
Appearance	Formation not uniform. Type I not 3-ply. Type II not single-ply. Type III not 2-ply. Not clean; presence of dirt, spots, holes, tears, shives or slivers. Does not compare favorably with comparison sample. Not embossed or creped. Not folded as specified.
Color	Not white.
Workmanship	Ragged, uneven edges.

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4.3.1.4 Examination of the end item for dimensional defects. The napkins selected for this examination shall be taken from the packages sampled under 4.3.1.3. Not more than two napkins shall be selected from any one package. Defects shall be scored as listed below.

Examine	Defect
Shortest dimension	Less than specified.
Area	Less than specified.

4.3.1.5 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) as follows:

Examination paragraph	Lot size units	Inspection level	AQL (defects per hundred units)
4.3.1.1	Shipping containers	S-2	2.5
4.3.1.2	Intermediate packages	S-2	NA
4.3.1.3	Sheets	S-4	2.5
4.3.1.4	Sheets	S-3	4.0

4.3.2 Testing of the end item. The end item shall be tested for the applicable characteristics in table II. The method of testing as listed in table II shall be followed. The lot size shall be expressed in shipping containers for each type and class of napkin. The sample unit for all tests shall be two packages. Not more than one sample unit, randomly selected, shall be drawn from any one shipping container. Five sample units shall be tested from each lot. The test report shall include all values upon which the results are based and there shall be no failure to meet the lot average or sample unit requirements. All samples shall be conditioned and tested in accordance with TAPPI method T402 except for dispensing operation.

TABLE II. Testing of the end item

Characteristics	Requirement paragraph	Test method	Results applicable to	Number determination per sample unit	Results reported numerically to nearest
Atmospheric conditions	4.3.2	T402	All testing except dispensing operation		
Basis weight	Table I	T410	Lot average	1	0.1 pound
Dry tensile strength	Table I	T404	Lot average	10 each direction	0.01 kg/15 mm
Wet tensile strength	Table I	T456	Lot average	10 each direction	0.01 kg/15 mm
Absorption time	Table I	T432	Lot average	10	1 second
Brightness	Table I	T452	Lot average	7	0.10 percent
Dispensing operation	3.7	4.4.1	Sample unit	1	Pass or fail

4.4 Test methods.

4.4.1 Dispensing. Load a dispenser conforming to RR-D-500 with 125 folded napkins (see 3.3.2 and 3.7). After the dispenser is loaded withdraw ten napkins in consecutive order. For the lot to be considered satisfactory, the napkins shall load and dispense without tearing or dispensing more than one napkin at a time.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

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5.1.1 Type I. One hundred or 125 napkins of type I shall be packaged in a box conforming to PPP-B-566 or PPP-B-676, wrapped in paper sleeves or banded with full width banding or packaged in a polyethylene bag to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.1.2 Type II, class 1 or 2. Not more than 250 napkins of type II, class 1 or 2 shall be packaged as specified in 5.1.1.

5.1.3 Type II, class 3. Two hundred and fifty, 500, 1000, or 2000 at the supplier's option of type II, class 3 napkins shall be packaged as specified in 5.1.1.

5.1.4 Type III. One hundred twenty-five type III napkins shall be packaged as specified in 5.1.1.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Two thousand napkins of type I, type III, or 10,000 napkins of type II, packaged as specified in 5.1, shall be packed in a snug-fitting box conforming to PPP-B-636, grade V3c or V3s. Each shipping container shall be provided with a liner conforming to MIL-L-10547, type I or II, grade B or C. The boxes shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Two thousand napkins of type I, type III, or 10,000 napkins of type II, packaged as specified in 5.1, shall be packed in a snug-fitting box conforming to PPP-B-636, class domestic, grade 275. Closure shall be in accordance with the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.1), 10,000 napkins of type II, packaged as specified in 5.1, shall be packed in a snug-fitting box conforming to PPP-B-1163, class 1, grade 275, type SWCFI, style PSC. The boxes shall be closed, waterproofed and reinforced in accordance with the appendix of that specification.

5.2.3 Level C. Napkins in quantities as specified (see 6.1), packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required in the procurement documents, marking of the interior packages and exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable (see 6.1).

6. NOTES

6.1 Order data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type and size desired (see 1.2.1, 3.3.1, 3.3.2 and 3.3.3).
(c) Dispenser (see 3.7).
(d) Level of packaging and packing required (see 5.1 and 5.2).
(e) Quantities required (see 5.1 and 5.2).
(f) Marking document required (see 5.3).
(g) When special level B is required (see 5.2.2.1).

6.2 Comparison sample. Comparison samples may be obtained upon application to the procuring agency issuing the invitation for bids.

6.3 The Department of Defense waives coordination on this specification.

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